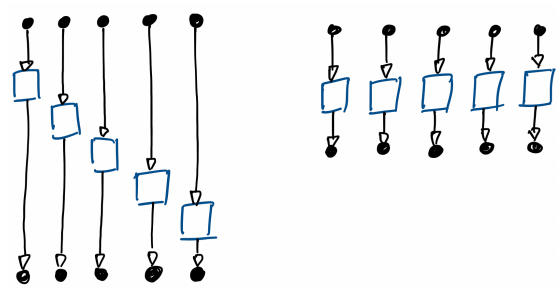
Parallel Pattern: Map

- replicates elemental function over every element of an index set
- map replaces iterations of independent loops
- elemental functions should not modify global data that other instances (iterations) depend on
- serial and parallel execution of the map pattern (elements of index set (black) and elemental functions (blue))

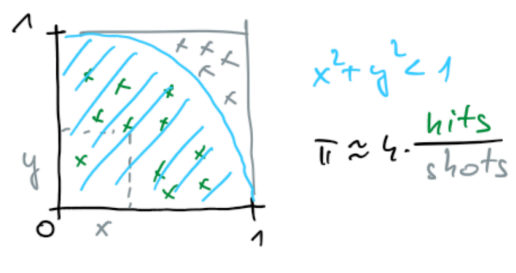


- examples:
  - gamma correction in images, color space conversions
  - Monte Carlo sampling, ray tracing
- map applies an elemental function to every element of a collection of data in parallel
  - elemental functions should have no side effects
  - no dependencies among elements
  - can be executed in any order
- embarrassingly parallel
  - one of the most efficient patterns
  - if you have many problems to solve, parallel solution can be as simple as running problems (serial code) on parallel execution nodes
- typically combined with other patterns
  - map does the basic computation, other patterns follow
  - reduction, gather = serial random read + map, scan
- scalable implementation of map
  - a lot of care is needed for best performance
  - threads
    - mandatory parallelism
    - separate thread for each element is not a good idea
  - tasks
    - optional parallelism
    - overhead due to synchronization at the start and end when elemental functions vary in the amount of work
- the map pattern is a basis for vectorization and parallelization
  - map is related to SIMD, SPMD, SIMT
  - can be expressed as a sequence of vector operations
- the parallel-for construct in programming languages
  - map is parallelization of the serial iteration pattern where iterations are independent
- if dependencies and side-effects are avoided, map is deterministic

Example

Monte Carlo π

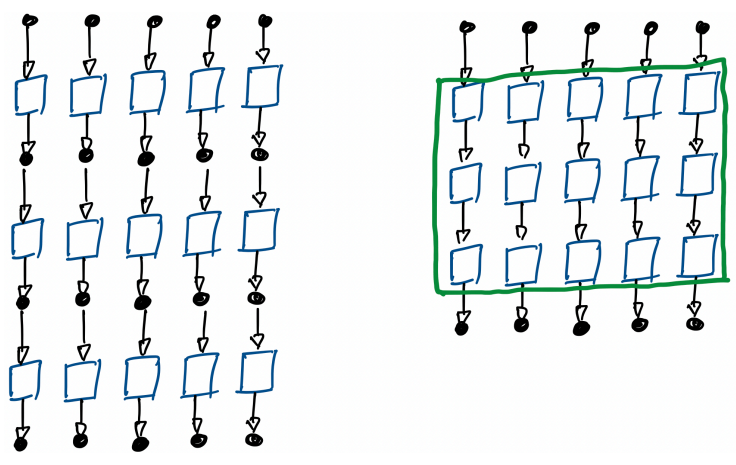
- Integration
  - random shooting to the square



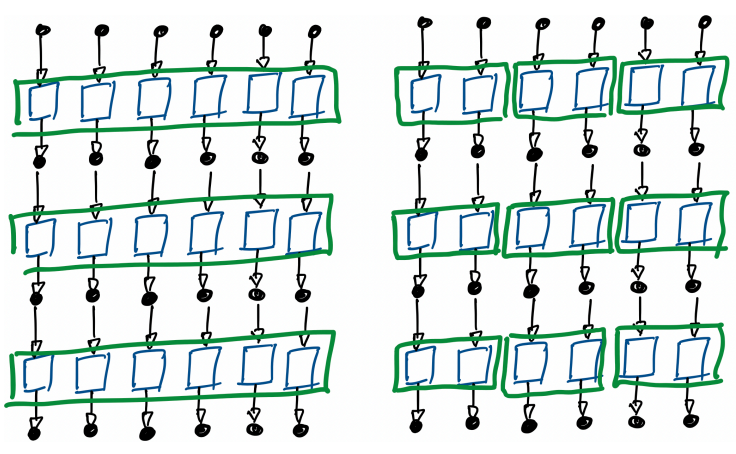
  - statistics of $shots$ and $hits$ ($shots$ inside the circular quadrant)
  - ratio $hits/shots$ is proportional to $\pi/4$
  - slow convergence, relative error is proportional to $shots^{-1/2}$
  - a basic unit of work that can be parallelized is one shot
    - a lot of overhead
    - better to combine several shots to one task
  - solution
    - sources: pimc.c and pisum.c
    - scripts:
      - pimc-1.sh
      - pimc-4.sh and pisum-4.sh

Code Fusion and Cache fusion

- code fusion
  - map of a sequence can avoid intermediate memory operations
  - intermediate results are written to registers
  - reduced memory bandwidth, cache and virtual memory issues
  - less synchronization at start/end of execution



- cache fusion
  - maps are split into tiles
  - each tile is executed sequentially on one core
  - tile should fit in cache, avoiding accesses to main memory
  - loop with predefined size inside a parallel section of a code
  - each map has the same chunk size



Patterns Related to Map

- stencil
  - access to neighbors to get inputs of elemental function
  - important is reading of data
    - data reuse
    - hardware specific for good results (cache size, GPU memory)
- workpile
  - work grows as it is consumed by map
  - could be implemented in OpenMP and OpenCL using explicit work queues
- divide-and-conquer
  - recursive division into smaller parallel subproblems until base is reached, which can be solved sequentially
  - combination of partition and map patterns
  - OpenMP: supported through tasking model
  - GPU programming: poor support for nested parallelism